

Topic: Selection Operation

Q. Explain the Selection operation in Relational Algebra. Write its syntax and provide examples demonstrating how it filters tuples.

Definition to Remember

The **S**
(tuples)

Selection Operation (σ)

* Operates on a **single table** (unary).

* Filter

2. Syntax of Selection

$\sigma_{\langle \text{predicate} \rangle}(\text{Relation_Name})$

* **Predicate:** Uses comparison operators ($=$, $\neq$, $<$, $\leq$, $>$, $\geq$) and logical operators (AND, OR, NOT).

3. Examples

Input Relation: EMPLOYEE

| Emp_ID | Name | Department | Salary |

| :--- | :-

Example 1: Simple Condition

Query:

Example 2: AND Condition (AND)

Query:

Example 3: OR Condition (OR)

Query:

Must-Write Points (for 10 marks)

1. Selec

Quick Recall

`sigma`